Get familiar with xv6's system calls, use them, and create new one!

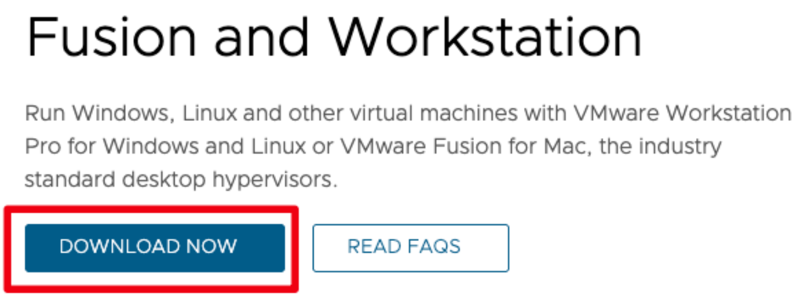
Environment Preparation

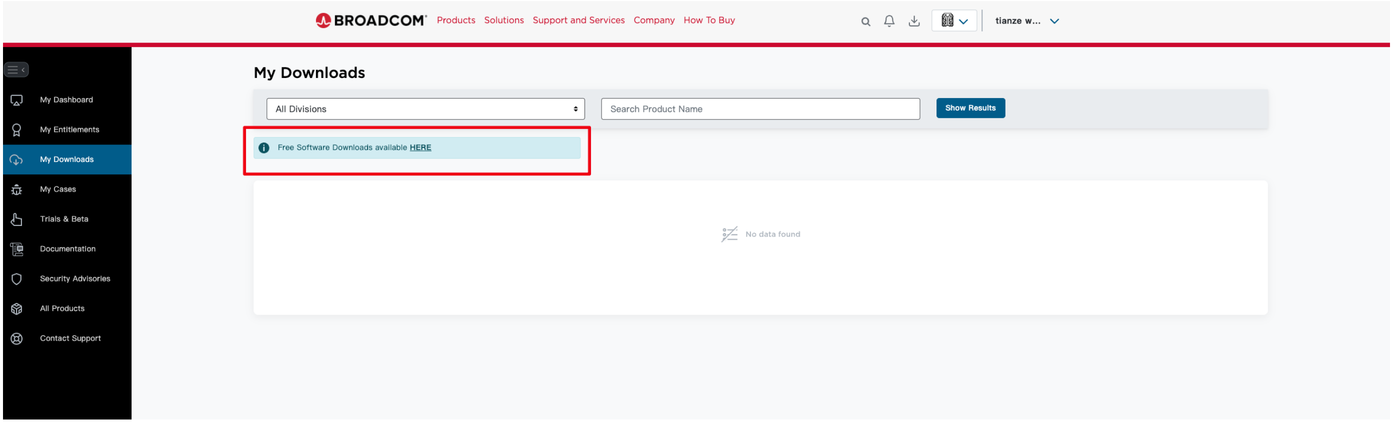
It's highly recommended to finish the labs under Linux environment. We have prepared a **Virtual Machine** image with Ubuntu 20.04 installed (https://releases.ubuntu.com/focal/).

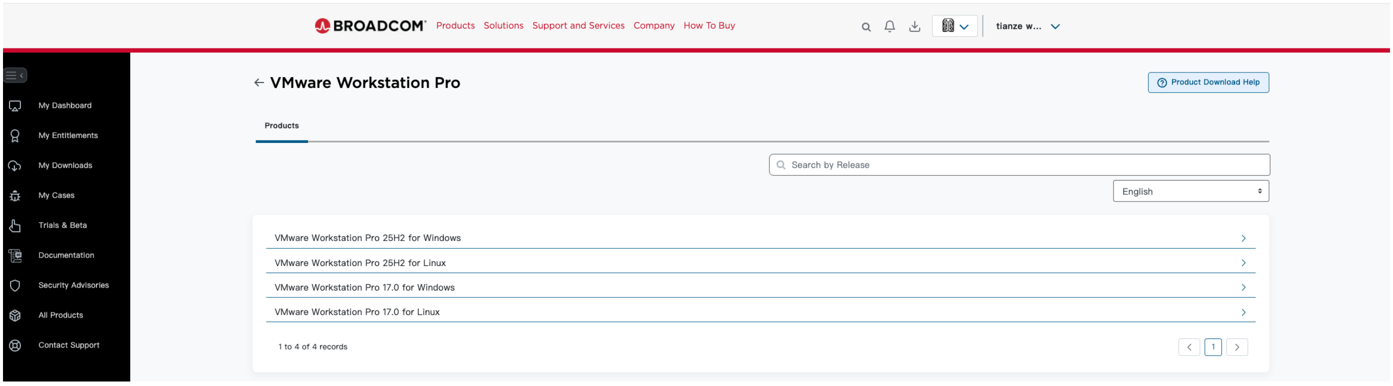
	Name	Last modified	Size	Description
	Parent Directory		-	
	SHA256SUMS	2026-03-10 17:14	297	
	SHA256SUMS.gpg	2026-03-10 17:14	833	
	ubuntu-20.04.6-desktop-amd64.iso	2023-03-16 15:58	4.1G	Desktop image for 64-bit PC (AMD64) computers (standard download)
	ubuntu-20.04.6-desktop-amd64.iso.torrent	2023-03-22 14:31	325K	Desktop image for 64-bit PC (AMD64) computers (BitTorrent download)
	ubuntu-20.04.6-desktop-amd64.iso.zsync	2023-03-22 14:31	8.1M	Desktop image for 64-bit PC (AMD64) computers (zsync metafile)
	ubuntu-20.04.6-desktop-amd64.list	2023-03-16 15:58	39K	Desktop image for 64-bit PC (AMD64) computers (file listing)
	ubuntu-20.04.6-desktop-amd64.manifest	2023-03-16 15:52	59K	Desktop image for 64-bit PC (AMD64) computers (contents of live filesystem)
	ubuntu-20.04.6-live-server-amd64.iso	2023-03-14 23:02	1.4G	Server install image for 64-bit PC (AMD64) computers (standard download)
	ubuntu-20.04.6-live-server-amd64.iso.torrent	2023-03-22 14:30	111K	Server install image for 64-bit PC (AMD64) computers (BitTorrent download)

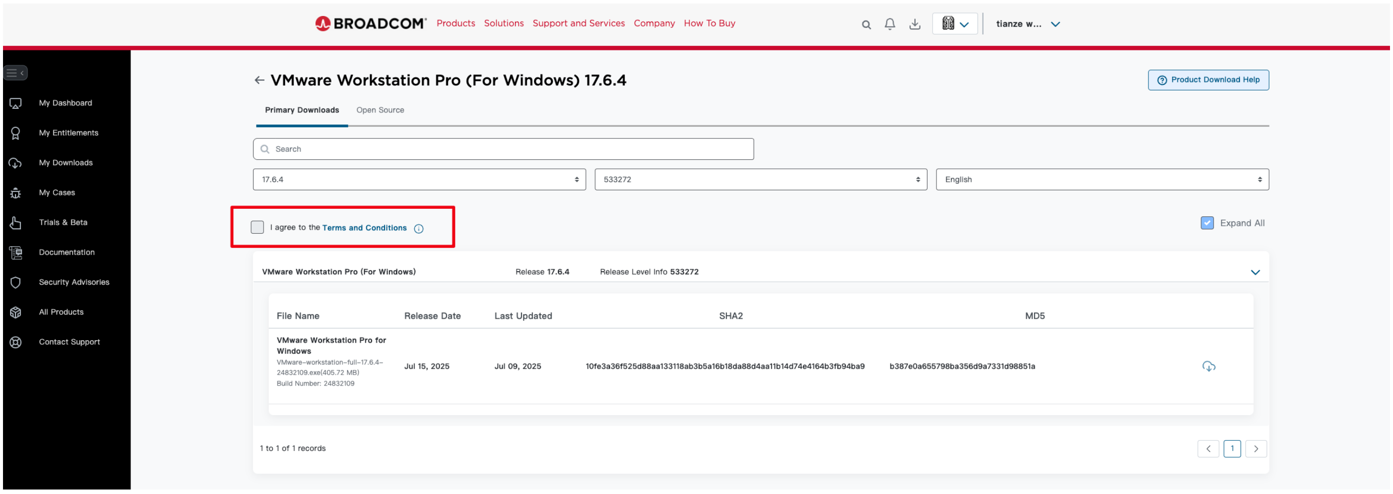
To use the VM image, a virtualization platform is required. For students who uses Windows OS, it is reommended to use VMWare (https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion). Choose workstation for Windows and fusion for MacOS.

(For students who uses Mac OS with Apple silicon, as our labs need to run under x86_64 architechture, you need to prepare a x86_64 environment.)









After loading the VM and login with the username and password above, the next step is to download the lab code using `git`:

$ git clone https://github.com/liyaoxuan/CS2953-2024.git
$ cd CS2953-2024 && make clean

Modify `conf/lab.mk` as `LAB=syscall`

Lab Tasks

sleep

> Implement the UNIX program `sleep` for xv6; your `sleep` should pause for a user-specified number of ticks. A tick is a notion of time defined by the xv6 kernel, namely the time between two interrupts from the timer chip. Your solution should be in the file `user/sleep.c`.

Run the program from the xv6 shell:

$ make qemu
...
init: starting sh
$ sleep 10
(nothing happens for a little while)
$

Some hints:

- Look at some of the other programs in `user/` (e.g., `user/echo.c`, `user/grep.c`, and `user/rm.c`) to see how you can obtain the command-line arguments passed to a program.
- If the user forgets to pass an argument, sleep should print an error message.
- The command-line argument is passed as a string; you can convert it to an integer using `atoi` (see user/ulib.c).
- Use the system call `sleep`.
- See `kernel/sysproc.c` for the xv6 kernel code that implements the `sleep` system call (look for `sys_sleep`), `user/user.h` for the C definition of `sleep` callable from a user program, and `user/usys.S` for the assembler code that jumps from user code into the kernel for `sleep`.
- `main` should call `exit(0)` when it is done.
- Add your `sleep` program to `UPROGS` in Makefile; once you've done that, `make qemu` will compile your program and you'll be able to run it from the xv6 shell.

pingpong

> Write a program that uses UNIX system calls to ''ping-pong'' a byte between two processes over a pair of pipes, one for each direction. The parent should send a byte to the child; the child should print ": received ping", where is its process ID, write the byte on the pipe to the parent, and exit; the parent should read the byte from the child, print ": received pong", and exit. Your solution should be in the file `user/pingpong.c`.

Run the program from the xv6 shell and it should produce the following output:

$ make qemu
...
init: starting sh
$ pingpong
4: received ping
3: received pong
$

Some hints:

- Use `pipe` to create a pipe.
- Use `fork` to create a child.
- Use `read` to read from a pipe, and `write` to write to a pipe.
- Use `getpid` to find the process ID of the calling process.
- Add the program to `UPROGS` in Makefile.
- User programs on xv6 have a limited set of library functions available to them. You can see the list in `user/user.h`; the source (other than for system calls) is in `user/ulib.c`, `user/printf.c`, and `user/umalloc.c`.

System call tracing

> In this assignment you will add a system call tracing feature that may help you when debugging later labs. You'll create a new `trace` system call that will control tracing. It should take one argument, an integer "mask", whose bits specify which system calls to trace. For example, to trace the fork system call, a program calls `trace(1 << SYS_fork)`, where `SYS_fork` is a syscall number from `kernel/syscall.h`. You have to modify the xv6 kernel to print out a line when each system call is about to return, if the system call's number is set in the mask.

> The line should contain the process id, the name of the system call and the return value; you don't need to print the system call arguments. The `trace` system call should enable tracing for the process that calls it and any children that it subsequently forks, but should not affect other processes.

We provide a `trace` user-level program that runs another program with tracing enabled (see `user/trace.c`). When you're done, you should see output like this:

```
$ trace 32 grep hello README
3: syscall read -> 1023
3: syscall read -> 966
3: syscall read -> 70
3: syscall read -> 0
$
$ trace 2147483647 grep hello README
4: syscall trace -> 0
4: syscall exec -> 3
4: syscall open -> 3
4: syscall read -> 1023
4: syscall read -> 966
4: syscall read -> 70
4: syscall read -> 0
4: syscall close -> 0
$
$ grep hello README
$
$ trace 2 usertests forkforkfork
usertests starting
test forkforkfork: 407: syscall fork -> 408
408: syscall fork -> 409
409: syscall fork -> 410
410: syscall fork -> 411
409: syscall fork -> 412
410: syscall fork -> 413
409: syscall fork -> 414
411: syscall fork -> 415
...
$
```

In the first example above, trace invokes grep tracing just the read system call. The 32 is `1<<SYS_read`. In the second example, trace runs grep while tracing all system calls; the 2147483647 has all 31 low bits set. In the third example, the program isn't traced, so no trace output is printed. In the fourth example, the fork system calls of all the descendants of the `forkforkfork` test in `usertests` are being traced. Your solution is correct if your program behaves as shown above (though the process IDs may be different).

Some hints:

- Add `$U/_trace` to UPROGS in Makefile
- Run make qemu and you will see that the compiler cannot compile `user/trace.c`, because the user-space stubs for the system call don't exist yet: add a prototype for the system call to `user/user.h`, a stub to `user/usys.pl`, and a syscall number to `kernel/syscall.h`. The Makefile invokes the perl script `user/usys.pl`, which produces `user/usys.S`, the actual system call stubs, which use the RISC-V `ecall` instruction to transition to the kernel. Once you fix the compilation issues, run trace 32 grep hello README; it will fail because you haven't implemented the system call in the kernel yet.
- Add a `sys_trace()` function in `kernel/sysproc.c` that implements the new system call by remembering its argument in a new variable in the `proc` structure (see `kernel/proc.h`). The functions to retrieve system call arguments from user space are in `kernel/syscall.c`, and you can see examples of their use in `kernel/sysproc.c`.
- Modify `fork()` (see `kernel/proc.c`) to copy the trace mask from the parent to the child process.
- Modify the `syscall()` function in `kernel/syscall.c` to print the trace output. You will need to add an array of syscall names to index into.

Sysinfo

> In this assignment you will add a system call, `sysinfo`, that collects information about the running system. The system call takes one argument: a pointer to a `struct sysinfo` (see `kernel/sysinfo.h`). The kernel should fill out the fields of this struct: the `freemem` field should be set to the number of bytes of free memory, and the `nproc` field should be set to the number of processes whose

> `state` is not `UNUSED`. We provide a test program `sysinfotest`; you pass this assignment if it prints "sysinfotest: OK".

Some hints:

- Add `$U/_sysinfotest` to UPROGS in Makefile
- Run `make qemu`; `user/sysinfotest.c` will fail to compile. Add the system call sysinfo, following the same steps as in the previous assignment. To declare the prototype for sysinfo() `in user/user.h` you need predeclare the existence of `struct sysinfo`:

  struct sysinfo;
  int sysinfo(struct sysinfo *);

  Once you fix the compilation issues, run `sysinfotest`; it will fail because you haven't implemented the system call in the kernel yet.
- sysinfo needs to copy a `struct sysinfo` back to user space; see `sys_fstat()` (`kernel/sysfile.c`) and `filestat()` (`kernel/file.c`) for examples of how to do that using `copyout()`.
- To collect the amount of free memory, add a function to `kernel/kalloc.c`
- To collect the number of processes, add a function to `kernel/proc.c`

Test

```
root@tzwang-server:~/CS2953-2024# ./grade-lab-util-syscall
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (1.0s)
== Test sleep, returns == sleep, returns: OK (1.5s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.0s)
== Test pingpong == pingpong: OK (1.0s)
== Test lab-util-syscall-report.txt ==
lab-util-syscall-report.txt: FAIL
    Cannot read lab-util-syscall-report.txt
== Test trace 32 grep == trace 32 grep: FAIL (1.0s)
    ...
         hart 1 starting
         init: starting sh
         $ trace 32 grep hello README
         exec trace failed
         $ qemu-system-riscv64: terminating on signal 15 from pid 2224286 (make)
    MISSING '^\d+: syscall read -> 1023'
    QEMU output saved to xv6.out.trace_32_grep
== Test trace all grep == trace all grep: FAIL (0.9s)
    ...
         hart 1 starting
         init: starting sh
         $ trace 2147483647 grep hello README
         exec trace failed
         $ qemu-system-riscv64: terminating on signal 15 from pid 2224321 (make)
    MISSING '^\d+: syscall trace -> 0'
    QEMU output saved to xv6.out.trace_all_grep
```

```
== Test trace nothing == trace nothing: OK (1.0s)
== Test trace children == trace children: FAIL (1.2s)
    ...
         hart 2 starting
         init: starting sh
         $ trace 2 usertests forkforkfork
         exec trace failed
         $ qemu-system-riscv64: terminating on signal 15 from pid 2224379 (make)
    MISSING '3: syscall fork -> 4'
    QEMU output saved to xv6.out.trace_children
== Test sysinfotest == sysinfotest: FAIL (0.9s)
    ...
         hart 2 starting
         init: starting sh
         $ sysinfotest
         exec sysinfotest failed
         $ qemu-system-riscv64: terminating on signal 15 from pid 2224408 (make)
    MISSING '^sysinfotest: OK'
    QEMU output saved to xv6.out.sysinfotest
Score: 45/80
```

In the example, 45 is your final score.

Optional challenge

- Print the system call arguments for traced system calls

Submit the Lab

For this lab, you need to submit:

- the source code of this lab.
- a report `lab-syscall-report.txt`

1. Validation

- Please run `make grade` to ensure that your code passes all of the tests
- Commit any modified source code before submission

2. Report

An report is needed, including errors you met when doing the lab, or some techniques you found that helps you to finish the lab better. You can use both **English** and **Chinese** to write the report.

3. Zip your files

1. Run `make clean` to remove excutable files.
2. Zip your code and report as a new file called `lab.zip`.

4. Submit

Upload `lab.zip` to canvas.